Synthetic Paper C: Section-Aware Chunking for Research Documents

Fixture ID SYN-C - Version 1.0 - Intended for project-level retrieval and paper comparison

TEST DATA DISCLOSURE: This paper is fictional. All study descriptions, sample sizes, measurements, and results below are invented solely to test MAIROS workflows. Do not cite as published research.

Abstract

This fictional study tests whether splitting research documents at section boundaries improves retrieval over fixed-character chunks. It compares a fixed 512-character strategy with a section-aware strategy on constructed paper excerpts. It supplies method, metric, result, and limitation fields for MAIROS comparison, hypotheses, experiment design, and literature-review testing.

Research question and hypothesis

RQ-C: Do section-aware chunks improve nDCG@5 over fixed-size chunks for questions about research methods and results? H-C: preserving section boundaries will improve nDCG@5, especially when a question targets a complete method or result passage. This remains a fictional hypothesis.

Methods

The fixture describes 60 synthetic research-paper excerpts. Fixed-size chunks use 512 characters with 64-character overlap. Section-aware chunking splits on headings and paragraphs, with a maximum target of 700 characters and a minimum of 120 characters. Both conditions use the same fictional query set, embedding model description, and ranking cutoff.

Dataset and measure

The invented primary measure is nDCG@5, averaged across the 60 constructed questions using fictional relevance labels. A secondary measure is multi-sentence answer-evidence recall: whether all sentences marked as needed for an answer appear within the top five passages. No actual model or benchmark was run.

Synthetic Paper C: Section-Aware Chunking for Research Documents — results and interpretation

TEST DATA DISCLOSURE: Every value on this page is invented illustrative data, not a real empirical finding.

Measure (invented)	Fixed 512-char chunks	Section-aware chunks
Mean nDCG@5	0.71	0.75
Complete evidence in top five	34 / 60 questions	40 / 60 questions
Median chunks per document	18	16

Illustrative interpretation

In this fictional fixture, section-aware chunks score higher on both invented retrieval measures and create fewer median chunks per document. These numbers are only fixture values for validating that MAIROS can index and retrieve content about chunking and can distinguish reported results from unsupported conclusions.

Limitations and open questions

All excerpts and relevance judgments are synthetic. The section parser is assumed to recognize headings and paragraphs; tables, equations, scanned pages, and irregular layouts are not evaluated. Open questions include how robust the gain is when headings are missing or misleading, how to select a maximum chunk length, and whether the same trade-off appears in multilingual or full-length real papers.

Comparison hooks

SYN-C concerns document segmentation; SYN-A concerns supported answer generation; SYN-B concerns graph-guided multi-hop retrieval. A reasonable test gap across the set is interaction: whether graph traversal still helps when chunk boundaries preserve section context. This is a test-design idea inferred from these three fixtures, not a claim about the research field.